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Historical imagery reveals cliffs as refugia for biodiversity in steep mountains

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ABSTRACT

Although protected areas are designed to halt human pressures and ensure the persistence of species and habitats of interest, they may experience land cover changes as a result of recovery from pre-protection disturbances. In this study, a two-step methodology was used to analyse the long-term landscape change and its potential impact on biodiversity in a steep southern European National Park. First, we used the WSL Monoplotting Tool software in conjunction with 18 centennial / current ground-based image pairs distributed over a vertical range of 2000 m to examine the fate of 1269 patches of seven land cover types. The use of a three-dimensional model allowed us to estimate real changes in an abrupt terrain where 27 % of its extent is hidden for classical or newer aerial imagery, and to assess the effect of topographic factors such as altitude, slope and perimeter-to-area ratio of the patch. The results showed an overall trend towards more advanced successional stages, with forests being the absolute winners at the expense of more open land cover types such as grasslands. Cliffs, on the other hand, showed the least change over the last century. Then, we analyzed 1108 plant inventories taken in these land cover types to assess how landscape change has potentially affected richness and the presence of plants of conservation interest (endemic, threatened, or species at the northern or southern limit of their distribution). The results suggest an overall loss of plants of interest over the last century, and highlight the conservation value of rocky habitats as refugia. Our analysis provides insights into the consequences of secondary succession of protected areas on biodiversity, and recommend some specific actions for an efficient management towards the most singular and vulnerable plant species.

1. Introduction

Mountains are widely recognized as hotspots of biodiversity (Körner, 2004; Rahbek et al., 2019), and excellent sentinels of global change (Jung et al., 2019; Spehn et al., 2010). They are often made up of semi-natural ecosystems where both natural and anthropogenic disturbance regimes (e.g., deforestation to create space for agriculture and grazing) coexist and have interacted for centuries. The complex interplay between these ecosystems characterised by high biodiversity and human use is undergoing changes in some high-income economies like Europe. This is due to a combination of factors, including the abandonment of traditional land uses following the significant depopulation of rural mountain territories and the designation of protected areas to mitigate anthropogenic

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pressures on biodiversity (Garbarino et al., 2014; Vicente-Serrano et al., 2019). These processes have led to changes in land cover, particularly the recovery of primary productivity (Ceaşu et al., 2015). Therefore, while protected areas may be less susceptible to habitat destruction, they are not entirely immune to changes in land cover (Hermoso et al., 2018). Given the heterogeneity of landscapes resulting from complex topography and anthropogenic activities, it is of interest to ascertain which environments undergo significant and minor transformation.

The natural recovery following the cessation or reduction in intensity of traditional uses can exert significant influence on biodiversity: some species will be favoured in habitats that undergo expansion, whilst others will be disadvantaged in habitats that undergo shrinkage. Moreover, the presence of stable environments during periods of landscape transformation has been shown to reveal areas that function as 'refugia', thereby facilitating the long-term persistence of typical inhabitants. Nevertheless, monitoring landscape dynamics does not automatically capture changes in biodiversity that might compromise the integrity of conservation areas' landscapes, because not all land cover types contribute similarly to the overall biodiversity. Therefore, it is necessary to assess the potential impact of changes in the abundance of different land cover types on key components of biodiversity. Detecting past long-term changes in land cover can thus constitute a first step for revealing their easiness for change, estimating risks to the biodiversity they shelter and prioritising the implementation of the most appropriate adaptive management.

Aerial photographs dating from the mid-nineteenth century, in conjunction with more recent remote sensing imagery, have extensively documented processes like forest fragmentation, the impact of fires, the transformation of landscapes following the abandonment of farmland in mountainous regions, or of habitats inhabited by imperilled species (Hu and Hu, 2020; Senf and Seidl, 2020; Eichenwald et al., 2020; Anderson et al., 2020). Nevertheless, the capacity to quantitatively track and interpret changes further back in time depends on the existence of older evidences, such as historical cartography (Tortora et al., 2015) or ground-based photography (oblique photographs). These historical pictures were typically taken by visitors and mountaineers for recreational purposes. They have since been used for descriptive visualisation and to quantitatively track changes in glaciers and coasts, or to characterise the effect of extreme events such as storms (Warrick et al., 2017). Not only do they extend back in time much further than other imagery sources (to the mid-18th century), they also enable the distinction of larger-scale changes that occurred during the first half of the nineteenth century in comparison to the second, a period during which many more information sources are available (e.g. aerial photographs, satellite imagery; Rhemtulla et al., 2002; Trant et al., 2020; Munteanu et al., 2024). Therefore, historical pictures represent the most accurate and precise baselines for ecosystem states, providing a reliable framework for assessing biodiversity changes. Furthermore, given that historical pictures are typically acquired from ground-based camera points at oblique view angles, they also encompass steep surfaces in abrupt terrains. The three-dimensional (3D) description of the landscape they provide, much richer of details than aerial photographs, facilitates the estimation of the true dimensions of steep areas and the contribution of features such as vertical cliffs, which are not discernible in vertical imagery.

When historical photographs are repeated decades later, the result is an image pair showing the same section of the landscape, making it possible to produce an almost perfect overlay. The recent advancement of new technologies to georeference oblique photographs to extract vector data (the WSL Monoplotting Tool; Bozzini et al., 2013) has enabled the accurate comparison of shifts in vegetation cover over time, provided ground control points (hereafter, GCPs) are used (Stockdale et al., 2015). Consequently, the analysis of historical images offers a much longer-term perspective for the study of steep terrains compared to the more sophisticated and precise remote sensing systems developed in the last four decades, becoming a unique tool to track long-term environmental changes in mountain systems (Sanseverino et al., 2016).

The present study adopts a two-step methodology to undertake integrated landscape and biodiversity change analysis in a mountain protected area characterised by considerable topographical complexity: the Ordesa and Monte Perdido National Park (OMPNP). Situated in the heart of the Pyrenean range, this area features Europe's highest calcareous massif mountain. Its topography is defined by five profound valleys, exhibiting a substantial altitudinal range (650–3355 m.a.s.l.). The lower areas of the valleys have historically been used for timber and agricultural purposes, while the upper areas for livestock grazing during summer. Following its designation as a National Park in 2018, the grazing practices in the alpine belt were permitted, although their intensity has diminished significantly over the past century (Fig. 3 in Gartzia et al., 2016). It is, therefore, a protected space worth monitoring to examine how natural recovering can affect biodiversity, taking into account that the pace and intensity of land cover changes is very heterogeneous due to the roughness of the area.

First, we repeat historical photography to analyse the land cover dynamics over the past one hundred years (early 20th century versus 2018–2022). Then, historical and actual oblique pictures were compared to identify land cover types that have undergone the most significant changes, as well as those that have exhibited the greatest resistance to change (i.e. refugia for biodiversity). The primary objective was to assess the advancement of the successional state of land cover, by answering to the following questions: 1) Which land cover types have been the winners and losers, and the most resistant to changes over the last century?; and 2) Which topographic and ecological factors have promoted changes versus resistance to changes of land cover?. We used a three-dimensional (3D) model instead of the traditional two-dimensional (2D) model to carry out this analysis, in order to take into account the actual size of the area under study.

The second step involves the exploration of plant diversity harboured by different land cover types, with the objective of estimating the extent to which landscape changes may affect different components. To this end, we collated and analyzed plant inventories performed in the study area over decades, and used the operative assumption that changes in the abundance of habitats translate into similar changes in the quantity and quality components of biodiversity they harbour. Biodiversity is a multifaceted concept that incorporates not only the quantitative component of richness, but also other aspects, such as phylogenetic or functional diversity, rarity, and specialisation (Gray et al., 2016; Pardo et al., 2017; Bonn and Gaston, 2005). Consequently, an exclusive focus on the quantitative component may prove inadequate for assessing components more closely related to the quality dimension (McGill et al., 2015).

Therefore, for the second objective, we compared biodiversity elements among land cover types that expanded, decreased or remained similar. This enabled to address the following questions: (3) What implications might have natural recovery had for biodiversity? and (4) Can any land cover type be considered as long-term refugia for biodiversity?

Reforestation and increase in primary productivity are well-documented trends in Europe and Mediterranean mountains in particular, irrespective of the protection status of the land (Vicente-Serrano et al., 2019; Matas-Granados et al., 2022; Álvarez-Martínez et al., 2014). Although revegetation processes can result from various situations, such as land irrigation, an intensive reforestation policy in degraded areas, or the natural recovery of areas after fires, the most widespread cause in the study area is the collapse of mountain agriculture and livestock activities, which transformed old cultivated fields on terraces and gentle slopes into areas covered by dense shrubs or forests. Consequently, we expect a general increase of the forest, particularly at lower, sunnier and less steeply areas, and a reduction of the rich flora of open habitats.

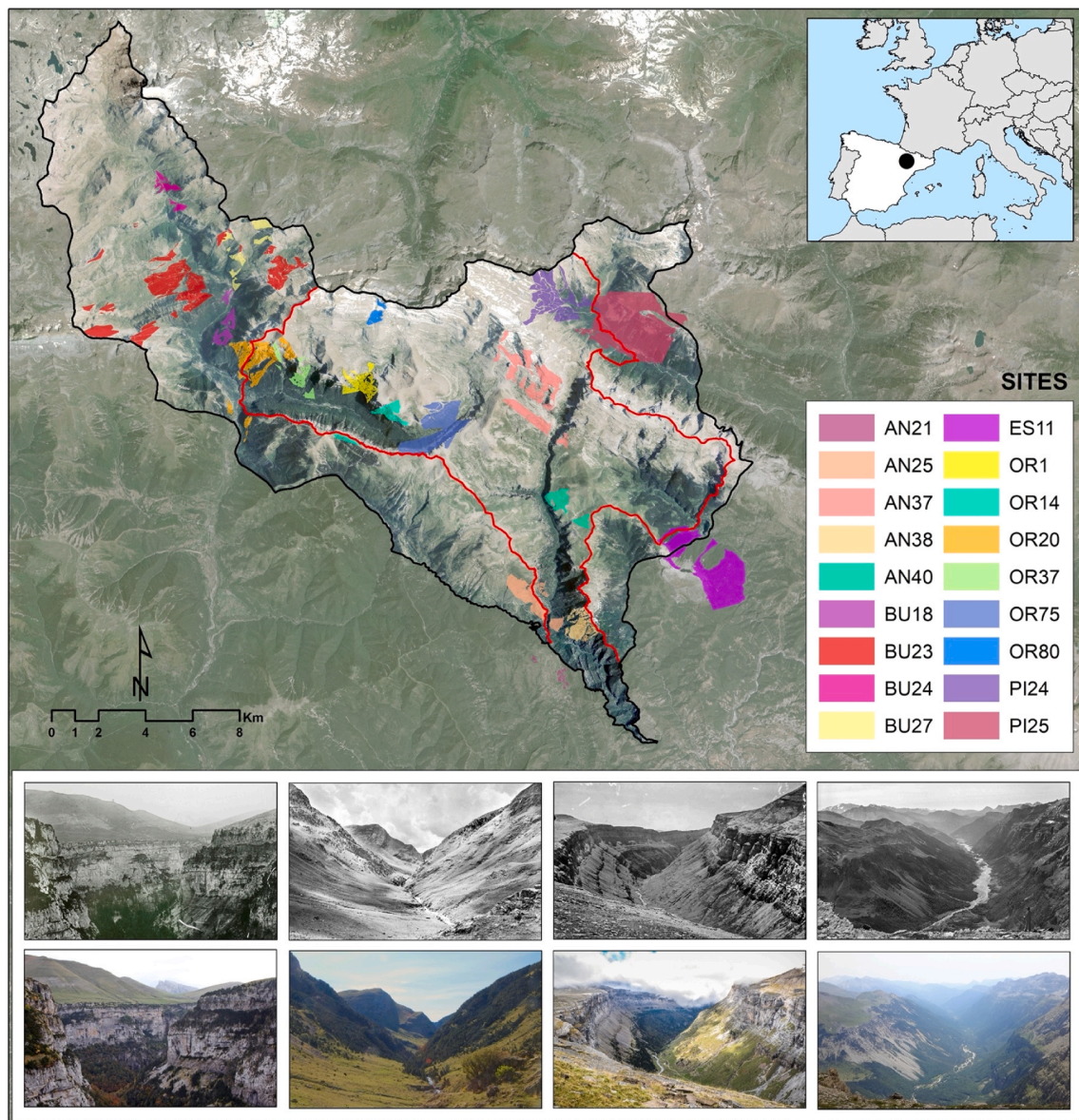


Fig. 1. Above: Location of the Ordesa and Monte Perdido National Park plus peripheral zone, and area covered by historical pictures (each colour corresponds to a different shot). Below: four pairs of centennial-actual pictures analysed to identify land cover changes. From left to right: AN40, BU24, OR75 and PI25.

2. Materials and methods

Ordesa and Monte Perdido National Park (OMPNP) is located in the southern central part of the Pyrenees. It is one of the oldest National Parks in Europe, declared in 1918, and then expanded in 1982. Although there are no people living in the protected area, and only a few small villages are located in the surrounding area, the OMPNP experienced a strong agropastoral use at the beginning of the 20th century, with potato and rye crops just below 2000 m a.s.l. The only activities allowed nowadays are traditional uses like summer grazing of livestock, mostly in the subalpine belt. Hunting, agriculture and timber harvesting are not allowed.

Land cover is highly heterogeneous (Benito-Alonso, 2019), with around 1300 vascular plants (almost 10 % of the whole European flora). The main habitats could be classified as follows taking into account their extent: subalpine grasslands (traditionally used for summer pasturing); rocky habitats (including rocky outcrops, screes and cliffs); coniferous forests dominated by *Abies alba*, *Pinus sylvestris* or *P. uncinata*; deciduous forests often dominated by *Fagus sylvatica*, and mixed ones; Mediterranean forests dominated by *Quercus ilex*; and shrublands.

2.1. Surface estimated from 2D and 3D terrain models

Following a standard estimation of a bidimensional (2D) cartographic model, the OMPNP extends over 34,805 ha (15,692 ha in the national park plus 19,113 ha in the peripheral area). However, for the purposes of this study, we have estimated the actual surface area based on a 5-m resolution Digital Terrain Model (3D) using the Add Surface Information tool from the 3D Analyst tools in ArcGIS Pro (<https://pro.arcgis.com/en/pro-app/latest/tool-reference/3d-analyst/add-surface-information.htm>).

2.2. Repeat photography

The primary information source for the historical pictures was black-and-white prints that were archived at public or private galleries and repositories (Fototeca de Huesca: <https://www.dphuesca.es/fototeca>; Centro Excursionista de Cataluña: <https://cec.cat/arxiufotografic/>), taken by naturalists and mountain climbers. The collection includes images dating back to the early 20th century (1900–1920), though the precise date of some is unknown and has been estimated based on the documented expeditions of notable travellers to the study area.

Following a thorough evaluation of the numerous available photographs (<https://paisajescentenarios.csic.es>), a final selection of 18 of high quality and free of snow were made, after accounting for a number of critical factors. Firstly, the geographical distribution

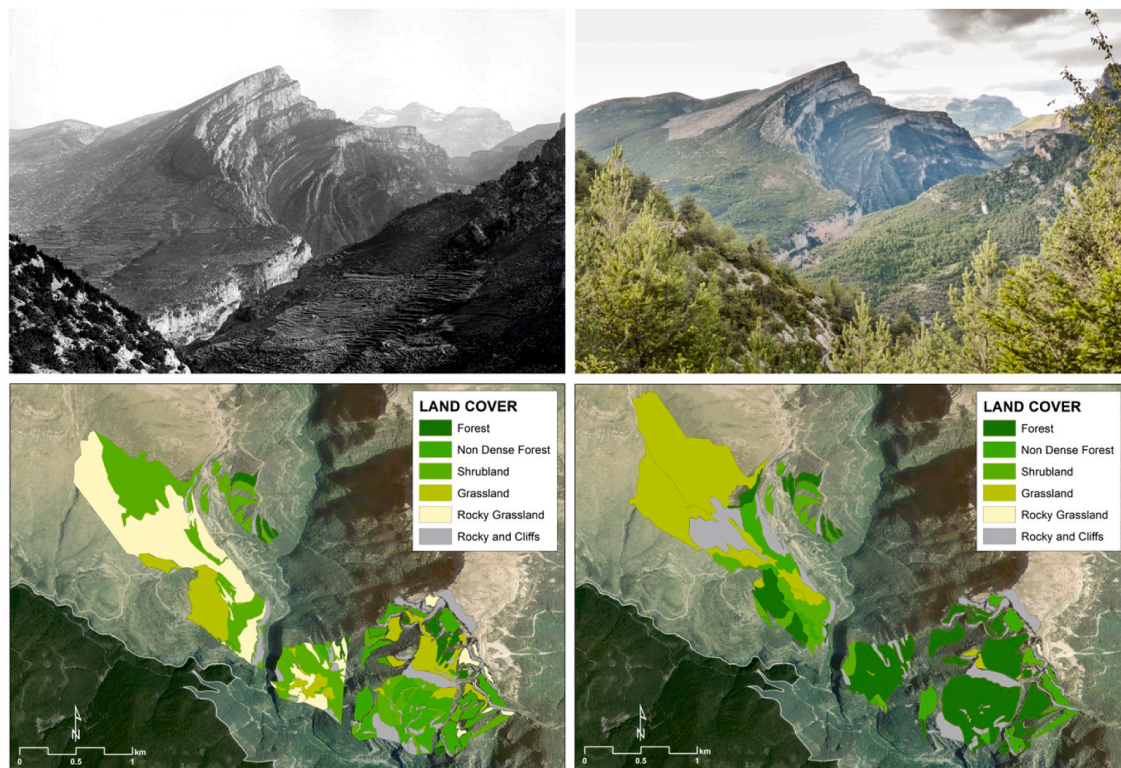


Fig. 2. Above: Historical (left) and recent (right) photographs of the same area in the Ordesa and Monte Perdido National Park, separated by about 100 years. Below: Patches of homogenous land cover delimited in both pictures with the use of the WSL Monoplotting tool.

across the five valleys in the OMPNP was maximised. Secondly, a wide landscape coverage was included to capture climatic and topographic gradients (from riverbeds in the lowlands to summits). The selection also encompassed the ease with which photographs could be captured in the same location, which proved to be a particularly challenging endeavour in some cases. This challenge arose due to the difficulty of locating the precise camera point or reshooting, which was hindered by the growth of vegetation. Selected pictures were repeated between 2018 and 2022, and pairs of images were digitised and saved in JPEG format, along with the relevant metadata (author, exact or approximate date, valley, rights of use...; see Fig. 1).

2.3. Delimitation and characterization of land cover patches

In order to track the fate of patches of homogeneous land cover types in old pictures we used the WSL Monoplotting Tool (Bozzini et al., 2013). This software facilitates the extraction of spatially referenced ecological information from oblique angle landscape photographs by calculating the actual coordinates of each point of the photograph using a Digital Terrain Model (DTM, 5 m of resolution in our case) and a mathematical model that simulates the geometry of the photographic applet: the collinearity condition (Price, 2010).

Firstly, GCPs were set in paired photographs (mean: 8.4, range: 5–16), from features that had been accurately identified (i.e. isolated trees, boulders). The georeferencing process yielded a median 3D error (uncertainty between the point in the photograph and the point on the map) of 13 m across all 36 images. Thereafter, areas of homogeneous cover in terms of both composition and structure were delineated manually, and the resulting polygons (patches) were assigned labels according to a general classification system of land cover types (see Fig. 2). A total of seven distinct classes could be distinguished in historical images: rocky (outcrops, scree, and cliffs when the slope was $>45^\circ$), grassland (dense, and rocky), shrubland, and forest (sparse when the density of trees was approximately $<20\%$, and dense). The characterisation of each patch in the historical images was undertaken by means of a number of topographic variables, derived from the DTM. These included the mean altitude, slope, aspect (in radians), total extent, length of the border, and ratio of border-to-area. Finally, patches were carefully examined in new pictures to either retain the classification if untransformed, or divide them into new patches if they underwent transformation into other land cover types.

In order to examine the overall restructuring of land cover types, a dataset of historical patches and associated topographic information was created. These variables included altitude, slope, aspect (decomposed in northern and eastern components from the $\sin$ and $\cos$ functions, respectively), area, and perimeter-to-area ratio. Each patch was then examined to determine if it remained the same or had to be reclassified or divided into smaller patches of different land cover in new pictures (to avoid misinterpretations due to manual delimitation of patches, reclassification was only applied to those whose change affected at least $\geq 10\%$ of the extension). Thus, we labelled each historical patch as 'changed' or 'unchanged', which enabled the estimation of the frequency of historical patches that transitioned into another class in the current landscape, and the exploration of which topographic and ecological variables exerted the most significant influence. A couple of transition matrices summarized directions and extension of change between pairs of land cover combinations between old and new photographs. The first matrix corresponded to the magnitude of changes, while the second to the probability of change of a patch in each land cover class.

A total of 1269 patches were delineated in historical pictures (patches smaller than 15 m^2 were removed, as they are most likely the result of mismatches between the borders of patches of old and new pictures). These patches covered 5116.6 ha (real area), corresponding to 11.6 % of the OMPNP and peripheral area across five distinct valleys. They encompassed a wide range of aspects, slope inclinations, and an extensive altitudinal gradient spanning more than 2000 m, from the valley bottoms to the alpine belt (Fig. 1).

2.4. Biodiversity: plant inventories

In order to assess potential changes in the overall richness, singularity and vulnerability of different land cover types that expanded, shrunk or remained stable over the one hundred years spanned between pictures, a search was conducted for plant inventories in the SIVIM platform (<http://www.sivim.info>; the Information of the Iberian and Macaronesian Information System (Font et al., 2012). The majority of inventories had been classified at the "alliance" level by expert botanists, which allowed for their integration into four major EUNIS habitats (European nature information system; the most comprehensive pan-European system for habitat identification, level 1): forest, shrubland, grassland and rocky. To ensure the classification, the FloraVeg.EU system (Mucina et al., 2016) was also used to classify inventories into alliances, with a 96 % success rate. This approach made it possible to classify inventories that lacked syntaxonomical information.

A total of 1108 plant inventories were analyzed, which included 22,196 plant records of 1086 vascular plants. This represents 73 %

Table 1

Analysis of Deviance. Binomial model to test the effect of different variables on the probability of change of 1269 patches distributed across the Ordesa and Monte Perdido National Park.

Variable	Df	Deviance Residual	Df Resid.	Dev	Prob (>Chi)	
			1268	1754.3		
Altitude	1	356.9	1267	1397.3	$<2.2\text{e}-16$	***
Slope	1	63.9	1266	1333.5	$1.32\text{e}-15$	***
Perimeter-to-area	1	4.3	1265	1329.1	0.03713	*
Land cover_historical	6	266.5	1259	1062.6	$<2.2\text{e}-16$	***

of the checklist of species recorded in the study area. The distribution of inventories across different land cover types is shown in Table 1. The frequency with which each type of inventory occurred exhibited a high degree of correspondence with the land cover occupancy of the analysed patches. The most prevalent land cover was rock (48.7 %), followed by grassland (24.2 %), forest (22.7 %), and shrubland (4.4 %).

A bibliographic review was conducted to ascertain whether each plant species was: 1) endemic either to the Pyrenean range where the Park is located (End_Pyr) or the whole Iberian Peninsula (END), 2) at the northern or southern limit of the species' distribution in the Pyrenees (LIM_Pyr), and 3) threatened according to the national Red List or regional catalogues (CAT). The integration of this information, namely "flora of interest", enabled the estimation of the contribution of each land cover type in terms of species richness, endemism (END), vulnerability (CAT), and the last frontier for species occurrence (LIM_Pyr). For this purpose, we calculated the frequency of plants in categories END, End_Pyr, LIM_Pyr and CAT for each inventory, and the mean ($\pm$ sd) frequency of END, End_Pyr, LIM_Pyr and CAT in each habitat. The number of inventories per habitat type, the mean number of taxa per inventory, along with information on richness and singularity of the flora surveyed, is presented in Table 1. Finally, the whole set of inventories was sorted and the frequency of inventories in the first quartil (Q1) for each plant category (End_Pyr, END, CAT, LIM_Pyr) was estimated in each habitat, as an indicator of their contribution to the quality component of plant diversity.

2.5. Data analysis

A logistic regression was conducted using patches defined on historical pictures as sampling units. The response variable was binary: changed / unchanged (logit function for the error term was used). The explanatory variables included the characteristics of the patches: altitude, patch area, perimeter-to-area ratio, slope, aspect and the land cover type (the seven classes defined previously).

The initial saturated model was reduced to include only variables that were found to be significant. As land cover type resulted to be significant, a *post hoc* test was conducted to ascertain which changes among the land cover classes were significantly different from others (Tukey contrasts). The deviance explained, and the predictive capacity of the model (AUC parameter, derived from a ROC curve), were then calculated. Finally, the model was checked to ensure it met all the necessary assumptions (e.g. graphics of residuals, no overdispersion).

In order to assess the potential impact of observed shifts in land cover on plant diversity, singularity and vulnerability, a comparison was made of the richness and frequency of endemics (END), threatened species (CAT), and those for which Pyrenean populations are at the edge of the distribution (LIM_Pyr) across land cover types. This was achieved by conducting contingency tables and Fisher exact tests, as well as pairwise *post-hoc* tests to analyze each land cover combination. Since we applied multiple hypothesis testing, *p* values were corrected using Benjamini-Hochberg FDR method for at a 5 % cut-off.

All analyses were conducted in R (R Core Team, 2024), packages DHARMA, ModEvA, multicomp, pROC, stats.

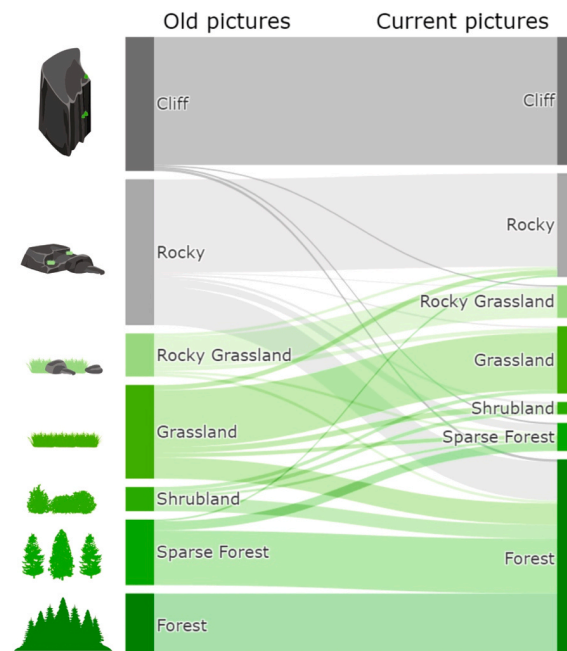


Fig. 3. Proportion of the analysed surface in the Ordesa and Monte Perdido National Park that has remained in the same land cover class or has been transferred to others over the last century. To facilitate visualisation, only changes larger than 10 ha were included.

3. Results

3.1. Land cover change

After accounting for topography (3D model), the real surface of the study area resulted to be 26.7 % larger than the estimated in bidimensional maps (44,111 ha in the 3D model vs 34,805 ha in the 2D model). The discrepancy between the 2D and 3D models illustrates the proportion of the surface that remains concealed when employing vertical vs oblique photography, and reveals the real extension available for plants able to thrive in steep surfaces. The images used in this study covered 5117 ha of the real surface (3D model), which corresponds to 11.6 % of the total surface. One third (33 %) of the analyzed surface changed over the last century, whereas 67 % did not change.

The analysis of repeat photographs revealed some general trends. On the one hand, land cover changes were mostly unidirectional, i.e. shifted towards more advanced successional stages (from herbaceous to woody species; Fig. 3), as 47 % of the 1269 old patches showed some transformation. Dense forests were the absolute winners, increasing considerably from all the remaining classes. Conversely, cliffs demonstrated remarkable stability over the past century, while grasslands, shrublands, and sparse forests exhibited the most significant shifts.

The logistic regression showed that the probability of a patch shifting to another land cover type significantly depended on the initial land cover type, as well as altitude (negatively), ratio perimeter-to-area of the patch (positively) and slope (negatively), being independent on patch area and aspect (Table 1). The deviance explained was 0.394, the probability of correctly classifying a patch that changed was 81 %, and the probability of correctly predicting a patch that did not change was 72 %. The model can be considered to have very satisfactory discriminant capacity (AUC) of 0.89. The *post hoc* test concluded that cliffs and dense forests were the land cover types whose fates differed most significantly from the rest of the classes, as they remained extraordinarily stable in comparison to the other types.

3.2. Potential effect of land cover changes on plant richness and species of conservation value

Regarding plant diversity, grasslands ranked first in terms of richness, followed by forests (Table 2). Rocky places ranked third in total richness and lowest in the mean of plants per inventory. However, the later ranked first in the mean occurrence of all kinds of interesting flora analysed: endemics (Pyrenees and Iberian Peninsula), species at limit of distribution, and threatened. Shrublands, the land cover that shrunk the most, were the least diverse, even though inventories showed the highest mean richness per inventory.

The frequency of endemics of the Pyrenean range, endemics of the Iberian Peninsula and species at limit of distribution in the Pyrenees significantly differed among the four habitats (Fisher test; $p = 0.0039$, 0.0087 , 0.010 respectively), whereas the proportion of threatened plants was very low in all the four habitats and did not differ between them ($p = 0.808$). Pairwise habitat comparisons concluded that the proportion of endemics (both of the Pyrenean range and the Iberian Peninsula) and plants at limit of distribution were significantly higher in rocky habitats than in other habitats (with the exception of the pair [rocky – grasslands] for species at limit of distribution). This highlights the key role of rocky areas for the long-term preservation of the most singular fraction of this mountain flora.

The selection of inventories in the first quartile (Q1) according to frequency of the flora of interest recorded concluded that rocky habitats shelter plant communities that include most endemics and species at limit of distribution. However, threatened species are more frequent in subalpine grasslands, which also shelter a high proportion of Pyrenean endemics (Fig. 4). Forest communities exhibited a comparatively low level of importance in relation to flora of interest, with the exception of some threatened species. Shrublands hardly contained plants of interest in terms of singularity or vulnerability.

Table 2

Frequency of inventories in the four EUNIS habitats, and abundance (Mean±sd) of different types of species of interest (End_Pyr, END, LIM_Pyr, CAT; see text for definition) per inventory.

	EUNIS Habitat description and code			
	Grasslands and lands dominated by forbs, mosses or lichens R	Heathlands, scrub and tundra S	Forests and other wooded land T	Inland habitats with no or little soil and mostly with sparse vegetation U
N (%) Inventories	433 (39.1)	36 (3.2)	179 (16.2)	460 (41.5)
Mean (sd) taxa	24.5 (10.7)	27.9 (14.5)	25.4 (11.5)	13.2 (7.7)
Total Taxa	765	384	645	586
N END_Pyr	23	10	13	33
Mean (sd) % END_Pyr	9.0 (6.9)	2.6 (3.6)	1.4 (0.3)	10.6 (9.5)
N END	54	24	38	62
Mean (sd) % END	9.0 (6.9)	5.1 (3.8)	2.6 (5.7)	19.8 (11.3)
N LIM_Pyr	144	59	103	128
Mean (sd) % LIM_Pyr	27.5 (18.7)	12.1 (8.8)	13.4 (9.9)	42.2 (20.0)
N CAT	8	2	6	4
Mean (sd) % CAT	0.1 (0.7)	0.1 (0.4)	0.4 (1.6)	0.5 (2.2)

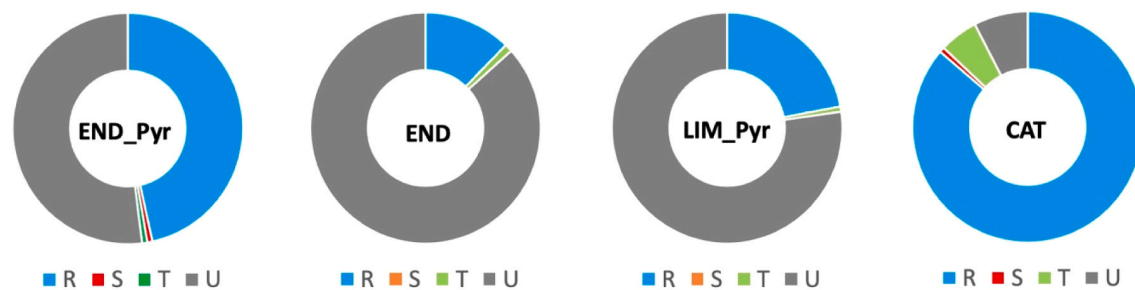


Fig. 4. Proportion of habitats that hold inventories ranked in the first quartile (Q1) of the sorted list for each kind of flora of interest: endemics of the Pyrenees (END_Pyr) or de Iberian Peninsula (END), species whose populations find in the Pyrenean range the northern or southern limit of distribution (LIM_Pyr), and plants listed in regional or national red lists or catalogued as threatened (CAT). R, S, T and U correspond to the four EUNIS habitats in which inventories were classified. R: Grasslands and lands dominated by forbs, mosses or lichens, S: Heathlands, scrub and tundra, T: Forests and other wooded land, U: Inland habitats with no or little soil and mostly with sparse vegetation.

4. Discussion

At a global scale, land cover is undergoing changes due to the increasing intensity of human activities, which have a detrimental effect on terrestrial biodiversity (Newbold et al., 2015). In this context, many studies focus on the effect of anthropogenic and natural disturbances on wildlife and how to design protected areas to reduce human pressures and halt biodiversity loss. The present study, however, confirmed that protected areas can also undergo a strong and deep in time land cover restructuring as a consequence of recovery after protection and farmland abandonment (Poulsen and Hoffman, 2015; Hermoso et al., 2018; Ameztegui et al., 2021). The use of centennial photographs and 3D models enabled us to estimate changes that have occurred in an abrupt terrain where an important extension is overlooked by more conventional aerial imagery techniques, and how such transformation varied considerably across land cover types. Combined with a large dataset of plant inventories collected in the area, this approach enabled us assessing the risk that natural revegetation in a protected area poses for plants of high conservation value, and highlighted the pivotal contribution of some inaccessible steep landforms (cliffs) to long-term biodiversity conservation.

4.1. Land cover changes from historical imagery

In the mountainous protected area under study, there was a significant transformation of open areas and sparse forest to closed-canopy forests, a common pattern observed in North American and European mountains, as well as the Mediterranean Basin (Rhemtulla et al., 2002; Sirami et al., 2010; Vacquie et al., 2015; Fortin et al., 2018; Stockdale et al., 2019; McCaffrey and Hopkinson, 2020; Bayle et al., 2025). Forests were the absolute winners, which are slowly recovering their “natural” place. Previous studies in the same area, based on aerial photographs taken in the 1950s and newer remote sensing methods have also shown significant landscape changes, such as the extension of woody vegetation and the displacement of forests to higher altitudes, with the concomitant decrease of grasslands (Gartzia et al., 2016; Ameztegui et al., 2021).

In order to better understand the overall trend of vegetation succession it was essential to include topographic information through the use of ground-based photography to reveal factors that promoted or slowed down changes to this mosaic landscape at real scale. Altitude significantly reduced the probability of transformation, which is logical given that: 1) areas at high altitude are more inaccessible to human intervention, 2) subalpine pastures are still grazed, albeit less intensively, and 3) plant life cycles are shorter at higher altitudes (lower recruitment, slower growth; Fillat et al., 2008), which translates into slower recovery in the upper parts. Other factors, such as the perimeter-to-patch area, a measure of exposure to colonization by shrubs and trees, also had a positive effect on the probability of land cover transformation, confirming that the National Park is transitioning towards a less diverse landscape (García et al., 2019). However, we did not find a significant effect of the aspect even though we expected more transformation in sunnier sides. Gartzia et al. (2016) also reported that woody plants spreading since the 1980s were linked to the westerly aspect in the grasslands of the Central Pyrenees, where the National Park is located. This lack of effect in our study could be a consequence of not covering the whole extension of the National Park. Finally, slope had a negative effect on the probability of changing, as gentle slopes can facilitate faster seed advance (Trant et al., 2020). Vertical rocky cliffs resulted to be the least transformed land cover type. The stability of cliffs has often been assumed (Larson et al., 2000), but rarely reported in geographic studies due to their hiddenness in vertical imagery. A comparison of historical photographs in the fymbos of the Cape region also revealed that rocky areas were the only places where woody vegetation remained when shrubs were dramatically reduced by burning (Marloth 1924, cited in Poulsen and Hoffman (2015)).

Repeat photography has some disadvantages. Firstly, very old pictures are not always easily available (mostly stored in private houses or public libraries), and secondly, these images rarely encompass the entire landscape. Historical oblique photographs taken from a human-eye point of view are common since the beginning of the photography in the mid-19th century and seem particularly suitable for the increasingly popular Citizen Science programs. An example worth mentioning is the Mountain Legacy Project (Sanseverino et al., 2016), which manages the world’s largest collection of historical mountain photographs, to explore changes in Canada’s mountain environment. New promising and faster techniques based on machine learning and neural network approaches will

soon increase its widespread use, provided that confusion between land cover classes is carefully avoided (Tricker et al., 2024; Morueta-Holme et al., 2024). Interesting new initiatives include an open API that allows researchers and developers to access and interact with an existing database of 200,000 3D georeferenced images (Ingensand et al., 2022). Another limitation is that images are not systematically repeated and often miss intermediate information about short-term disturbances. Therefore, they seem particularly suited to areas where long-term, slow and deterministic changes dominate.

Nevertheless, although the comparative analysis of historical images has some limitations compared to more modern technologies, this technique can provide quantitative estimates of vegetation dynamics from a unique perspective, providing the only evidence available for assessing long-term vegetation change on steep terrains. The strength of repeat photography also lies in the length of time it covers (in our study case from the highest rural density at the beginning of the 20th century till nowadays) and the perspective of nature it captures, which can complement the more sophisticated and costly contemporary techniques (Depauw et al., 2022; Morueta-Holme et al., 2024). The advancement of new technologies over the last five decades has probably hindered the use of repeat photography and underestimated its potential for assessing landscape change, although it does provide information on changes that could not otherwise be achieved (Moore et al., 2016; Ingensand et al., 2022; Depauw et al., 2022; Morueta-Holme et al., 2024). Fortin et al. (2018) examined the discrepancy in landscape changes between the ground-based imagery and the newer Landsat imagery, and found that both estimated proportions of change in a similar way, particularly for dominant land cover classes. However, the oblique imagery was better at capturing both narrow and small areas such as wetlands, due to the classes associated with steep terrain and the more appropriate angle of incidence. As stated in Bayr (2021), although oblique photographs cannot compete with the spatial accuracy of top-view imagery, they offer distinct advantages, including temporal depth and their capacity to communicate landscape change in a clear and comprehensible manner (“a picture is worth a thousand words”, Hammond et al. 2020).

4.2. Biodiversity changes

Land cover change leads to either an increase or decrease in landscape heterogeneity, which in turn affects many aspects of the ecosystems, such as species richness (Costanza et al., 2011). As more diverse landscapes provide a wider range of habitats for a greater diversity of species, the resulting landscape heterogeneity will determine an increase or decrease in the amount and quality of biodiversity supported. However, current knowledge of how changing spatial attributes affect biodiversity is limited by the scarcity of studies focusing on how this affects species persistence or dynamics (Sitzia et al., 2010).

The densification or expansion of forest we found is one of the dominant processes in Europe (Sitzia et al., 2010; Zanden et al., 2017) which often leads to a decrease in landscape heterogeneity (Sychrová et al. 2024). According to the plant inventories conducted in forests we would predict that forest expansion would have a high contribution to richness. However, past land uses can exert long-lasting effects on the structure and functioning of restored landscapes (Munteanu et al., 2015), so it is not straightforward that their potential contribution occurs as soon as they colonize a different habitat. Nevertheless, despite the fact that forests do not shelter a large fraction of flora of interest, they give protection to many species, and stand out in terms of another components of biodiversity that rarely are taken into account: phylogenetic diversity and evolutionary distinctivity (García et al., 2022; Miranda-Cebrián et al., 2022). Forest also constitute classical climate refugia that slow down the response of plant diversity to contemporary climate warming (de Frenne et al., 2013; Zellweger et al., 2020; Hoffrén et al., 2022).

Forest expansion, however, occurred at the expense of more open land cover types, such as grasslands, supporting previous results obtained through aerial imagery in this and other mountain National Parks in the Iberian Peninsula (García et al., 2019; Ameztegui et al., 2021). As grasslands harbour a high proportion of threatened flora and a considerable proportion of endemics and species at their limit of distribution, current management should focus on maintaining landscape diversity by limiting secondary succession within open areas (Garbarino et al., 2014). Sirami et al. (2010) found in a natural Mediterranean area in south France that land cover changes had a negative impact on long-term trends of birds and reptiles, especially rare and endemic species and those with an unfavourable conservation status. Niemelä and Baur (1998) also found in the Alps that shrub encroachment and forest expansion were beginning to cause concern and considered a major problem for plant and animal conservation.

Finally, rocky habitats, which are overlooked in 2D cartographic maps, emerge not only as a more widespread land cover than recorded, but as being of great uniqueness and conservation value due to the high proportion of endemics and other species for which the Pyrenees constitute the last northern or southern frontier (Tejero et al., 2017; García et al., 2020). The role of cliffs and other rocky substrates as refugia for cold-adapted plants in the deep past has been highlighted too, on the basis that north facing cliffs allow them to grow below the altitudinal forest limit (Birks and Willis, 2008). Cliffs also offer a variety of microhabitats, such as small landings with developed soil on the rock faces or megafossils at the base, thus providing small refuges for forest and grassland species. They can therefore act as reservoirs of biodiversity when ecological conditions change.

The importance of considering the effect of relief in conservation studies is clearly paramount for species associated with steep terrains, as this approximation is closer to the true landscape (Galindo-Cruz and Rojas-Soto, 2023). Because of their high stability during landscape transformation and low likelihood of being affected by human disturbance in protected areas, rocky habitats play the role of refugia for the organisms that can thrive there (Larson et al., 2000; Kwiatkowski, 2021). Our results clearly indicate that they should be considered as an important long-term conservation target (March-Salas et al., 2023), especially in the Mediterranean Basin, home to many endemic plant species (Cowling et al., 1996).

5. Conclusions and take-home messages

Protected areas are often designed to preserve habitats and species that are unique and scarce or highly vulnerable from

anthropogenic pressures. However, since landscapes are not static, protected areas require the periodic assessment of the effects of global change drivers on biodiversity and the processes that maintain it. Our study combined the analysis of land cover dynamics over the last century in a topographically complex National Park, with the occurrence of plant species in different land cover types. The comparison of oblique historical and current photographs enabled the quantification of three-dimensional changes based on the oldest visual evidence, providing the best testimony of centennial changes at the landscape scale. The results revealed that natural recovery after protection does not guarantee an integral preservation of biodiversity since the expansion of forests occurs at the expense of grasslands, which increases the probability of extirpation of endemics or threatened plant populations. They also highlighted the role of cliffs, hidden in two-dimensional maps, as refugia for the long-term persistence of a subset of plants of high conservation value.

In view of the observed trends in land cover, it is imperative that certain management actions are implemented to ensure the conservation of the park's most unique and vulnerable plant species. Ideally, the traditional land use that has created the centuries-old landscape mosaic should be maintained. However, it appears impracticable to reinstate customary practices in a park that has undergone significant depopulation and receives hundreds of thousands of visitors annually. Conversely, the reduction of forest density and the deliberate creation of artificial forest openings are recognised as evidence-based management practices that support multiple ecosystem services. A reduction in forest cover has been demonstrated to increase streamflow and water yield, whilst concomitantly reducing the water stress experienced by remaining trees (Bosch and Hewlett, 1982; del Campo et al., 2022). Limiting tree expansion could also be an effective way to reduce the risk of fuelling fires in areas where carbon storage is increasing rapidly. Prescribed burning has been justified in southern Europe by the need to manage fire-prone vegetation types and maintain cultural landscapes (Fernandes et al., 2013 and references therein). This practice seems to have neutral or positive effects on soils and biodiversity, in contrast to wildfires, which can be extremely damaging. Its use in the Pyrenees has facilitated the revival of best-practice traditional fire use within a regulated framework. However, in order to assess the effectiveness of any proposed forest management strategy, it is essential to use site-specific eco-hydrological models, which must take into account a variety of factors, including biodiversity, carbon, fire risk and nutrient balance.

The shadowing of the increasingly denser forests diminish the quality of the habitats for light-demanding species that are unable to tolerate full shade. Forest clearings increase heterogeneity and ecotones and enhance biodiversity because tree removal increases solar energy and precipitation reaching the forest floor. A recent review concluded that forest thinning significantly increased shrub diversity by 28 % and herb diversity by 24 %, the coniferous forest being more favorable for understory improvement than the broadleaved forest (Li et al., 2020). The control of tree density and the prevention of their edges approaching the cliffs and screes would also be of particular interest in the preservation of the integrity of habitats that are home to many endemic plants. Their effectiveness has been demonstrated by experimental forestry practices in the Swiss Jura mountains (Müller et al., 2006).

Understanding how the present landscape came to be after disturbances ceased requires the best and deepest quality baseline measurements of ecosystem states. Repeated historical photography can provide such a baseline and has allowed us to confirm that protection alone does not guarantee optimal conservation outcomes. A comprehensive approach to past changes in biodiversity, as employed in this study, has the potential to serve as a simple tool to guide active management and conservation planning in increasingly depopulated mountain landscapes. Citizen science initiatives, such as the one utilised in this study (<https://paisajescentenarios.csic.es>), can contribute significantly to the monitoring and conservation of cliffs, given their aesthetic appeal. The integration of public engagement would facilitate not only the collection of data but also raise awareness of the existence of refugia and their transcendental role for biodiversity in a transforming world. In addition, in a technologically advanced world, the implementation of sophisticated monitoring systems can facilitate the future tracking of not only the geographical distribution of land cover patches at broad scale, but also the productivity of forests or grasslands at metric scale, and even the fate of individuals of threatened plants in inaccessible cliffs (e.g., drones; Harrison et al., 2024; Nyberg et al., 2024). The combination of past trends and contemporary monitoring systems, in conjunction with stakeholder and public participation, will serve as a powerful instrument to enhance future biodiversity persistence in a changing world.

CRediT authorship contribution statement

María B. García: Writing – review & editing, Writing – original draft, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Carlos Díaz-Gil:** Writing – review & editing, Visualization, Resources, Investigation, Formal analysis. **Enrique Aparicio:** Resources, Investigation. **Marta Quintana-Buil:** Resources, Investigation. **Manuel Pizarro:** Visualization, Methodology. **Daniel Gómez:** Resources. **Xavier Font:** Resources, Data curation.

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Declaration of Competing Interest

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Data availability

Data will be made available on request.

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